Finding Optimal Path Planning Method for Building Navigation in BIM

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Building's spatial structure described in a BIM model can be used for retrieving the information required for determining the shortest path within the building. The matrix method and the visibility method are the two main ways of dividing space into prime factors. Both are widely used to find the shortest path. In order to compare the performance of both methods, several tests were carried out with various versions of the floorplan modification (room area, the surface of internal walls, distance between the entrance, and exit in a straight line and within the boundary marked by walls of the maze). The results revealed significant differences between the visibility graph and the matrix method.

Keywords: *BIM, Building information modeling, Facility management, FM, Routing in building*

INTRODUCTION

Navigating inside a building differs from finding the road between points A and B on a street map and is similar to finding the shortest route in an open space. While the street grid is a ready-made material to be processed by a search algorithm, the interior of the building is a bigger challenge as it is a set of rooms that are not linear and that are often irregular. The BIM model, as a carrier of data about the spatial structure of a building, is an ideal source of information allowing to determine the shortest route between various points in the facility. A script must translate a BIM-geometry to a graph build of points connected with lines.

Creating an effective building navigation tool is an example of the use of a BIM model throughout the entire life cycle of a building beyond the most obvious "design and build" (Volk, Stengel and Schultmann, 2014). Thanks to automating many decision-making processes and increasing work efficiency, it can be successfully used in tasks related to the ongoing maintenance of the building (Chen et al., 2018).

The work aims to find the optimal solution for determining the shortest route between points in the building. A model made in Revit is a source of spatial information and a script created in Dynamo is used to determine the shortest route. The lengths of the mapped routes and the times needed to perform the calculations are recorded and compared. The research is conducted in rooms (labyrinths) with varying degrees of complexity to find the major factors influencing the performance of the calculations and perform their effectiveness.

LITERATURE REVIEW

Decomposing an open space can be achieved in several ways. For example, transformations of space's boundaries may be used to investigate it. Methods based on a visibility graph (Masehian and Amin-Naseri, 2004; Krisp, Liu and Berger, 2010) and a skeleton method (Haunert and Sester, 2008) are included in this group. Another choice is to use a matrix method, which is built on a matrix of points connected to each other either by rectangular grids or grids expanded by the Delaunay triangulation and allows for Spider method routing (Hahmann et al., 2018). According to the research presented in (Hahmann et al., 2018), none of the algorithms listed above is universal, but the visibility graph and the Spider algorithm are the most promising. As a result, only these two methods will be considered further.

As the matrix method is based on a set of points connected with each other, there is some degree of user interference in this method, as the researcher typically assumes the density of the grid in comparison to the evaluated space (Graser, 2016; Hahmann et al., 2018). The researcher evaluates the size of the space and chooses between a "dense grid" and a "loose grid" at choice. There were not found references describing attempts to test this method in objective criteria and to find a balance between the advantages and disadvantages of the grid-based matrix approach. It is believed that a higher grid density improves path mapping accuracy (Chen et al., 2018). Although this is right, increasing the grid density is not necessarily advantageous in terms of maximizing the planning process since it increases the computation time (Rüppel, Abolghasemzadeh and Stübbe, 2010). This approach has certain primary benefits, such as interpolating points within the open space and generating routes with lengths similar to the shortest paths, as well as the biggest drawback of accurate measurement - an improvement in calculation time (Hahmann et al., 2018).

In the second method, using the visibility graph, a researcher cannot add their subjective attitude, as space is analyzed in the context of its boundaries, more specifically its corners, which become nodal points. Then, from each point, the maximum number of lines is drawn to the corners that are "visible" from a given place. Visible, that is, a straight line between the points can be led and this line does not intersect with any elements of a building structure.

As a result of both transformations, space becomes a mesh, and an algorithm can calculate the route between the points. In this paper, both the abovementioned methods are compared in terms of efficiency and applicability together with the BIM model.

RESEARCH TOOL

The analysis was carried out for a maze-shaped space with just one entry and exit. The room's walls were entered at random. The study was carried out in Autodesk Revit 2018 with Dynamo 2.0.2.6833, an advanced add-on for visual programming. In Dynamo, the experiments were made using the same script. Since Revit is one of the most common BIM programs and is closely linked to Dynamo, a visual programming application, this software environment was chosen. In Revit 2020 the new tool "Path of travel" was added. This tool finds the way between two given points on the selected level. This tool uses the visibility method to find the way, although it cannot operate between levels so it is a closed internal tool suitable rather for calculating evacuation times during the design phase than building navigation.

Overall, Revit + Dynamo proved to be a useful tool for performing the study. The computer was not used for any other purposes during the calculations, and all analyses could be considered to be conducted in equal conditions. The Dynamo script used in the analysis extracted information from the BIM model about the room, walls (graph edges), and doors (the entry and exit points).

The obtained input data are transformed depending on the chosen research method. As mentioned, the method based on the visibility graph does not require the user to enter additional information. The visibility method provides a graph created by

connecting all the vertices in the room. Vertices are generated in Dynamo by converting the boundaries of the room to a polygon. The figure presented below (Fig. 1) shows a visibility graph created in a labyrinth which is one of the case studies in the research.

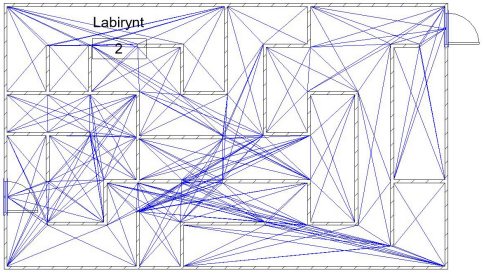


Figure 1
Floorplan of the maze used in Series 1-4 with the visibility graph on it

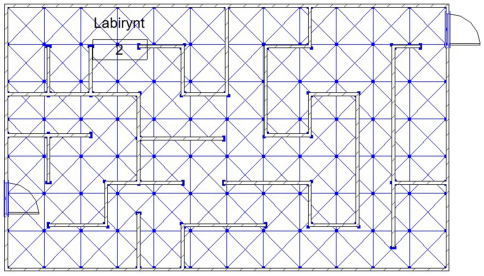


Figure 2
Floorplan of the maze used in Series 1-4 with the matrix grid on it

Table 1
Description scheme for each test

The matrix method can change the density of the grid of points. The number of segments along the x and y axes was the rounded down result of dividing the length and width of the room by the user-defined factor x representing the "grid density" (Fig. 2). The densities were lowered by 10 from 100 to 10.

In both methods, it was assumed to find the shortest way connecting points. It should not be associated with a path planned for humans or independent devices. In order to prepare to polygon for planning routes for things with a given size, the border of the room would need to be offset.

RESEARCH

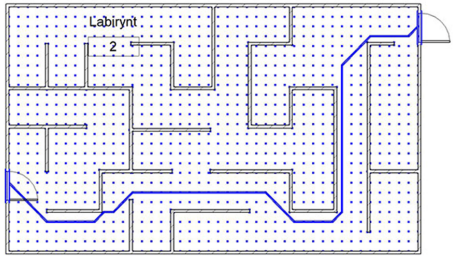
The maze variants slightly differed from one another. Each of 1-4 series of experiments was conducted on a maze with the same wall structure but different door locations (Tab. 2-5). In the first sequence, the doors were on both sides of the labyrinth (Tab. 2). In contrast to the first test, the doors in the second series were closer to each other (length in a straight line), but at a comparable distance of the maze passage (Tab. 3). In a straight line, the doors in the third series were identical to the doors in the second series, but the path to the passage was almost twice as short as in Series 2 (Tab. 4). The Series 4 doors were designed to create the longest passageway possible (Tab. 5). The value of the coefficient x (grid density), Dynamo's script execution time, and the route length were all reported for each sequence. Table 1 displays the classification scheme.

All 4 layouts are presented below with the solution for the matrix method for the grid x=30.

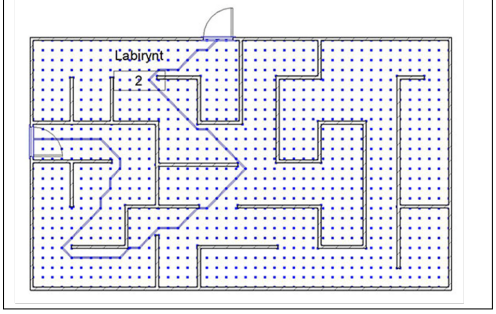
Series No.	Grid density – x [cm]	Calculation time [mm:ss]	Distance [cm]
Figure - Floorplan of the maze with a grid and a marked exit route			

Table 2
Results obtained for x=30 for Series 1

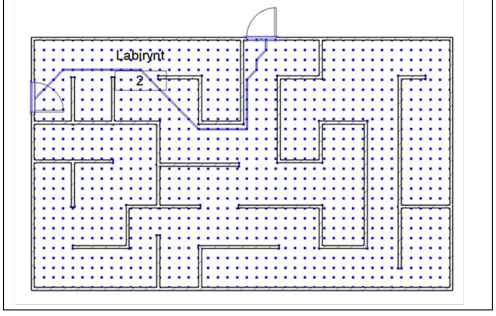
1	30	00:45.6	1943



2	30	00:46.3	2022



3	30	00:39.4	1076



4	30	00:42.1	2512

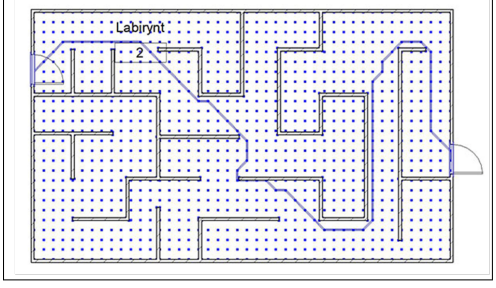


Table 3
Results obtained for x=30 for Series 2

The charts below present the data obtained for each of the 10 variants of the variable x (Fig. 3-6). In the matrix method, the general trend of reducing route length while increasing mesh density was reported.

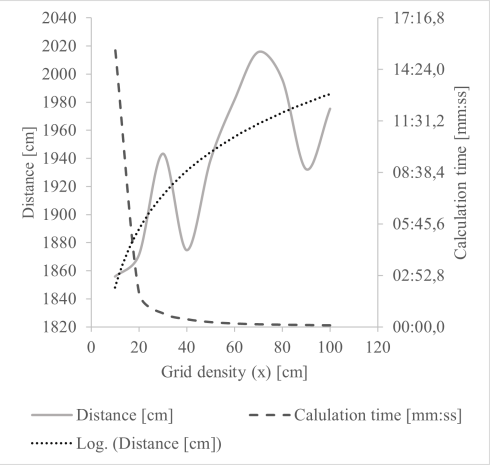


Figure 3
Visual comparison of results for Series 1

Table 4
Results obtained for x=30 for Series 3

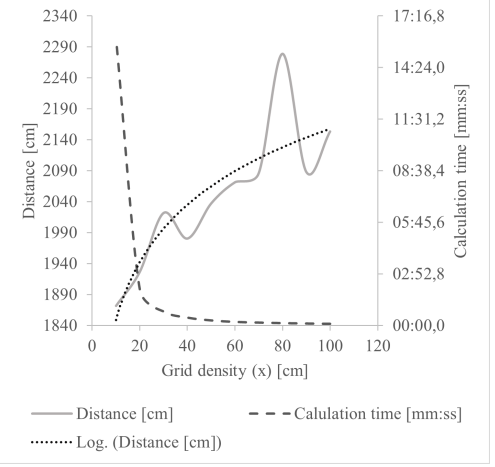


Figure 4
Visual comparison of results for Series 2

Table 5
Results obtained for x=30 for Series 4

Figure 5
Visual comparison of results for Series 3

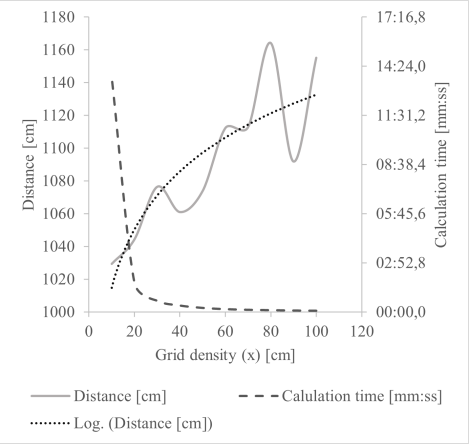
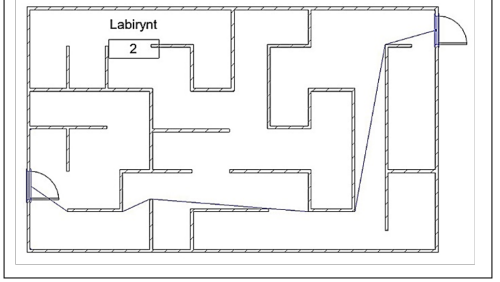


Table 6
Results obtained for visibility graph for Series 1

1	n/a	00:01,1	1778



Although the route with the shortest distance is given by the densest mesh, the time of calculation grows significantly after exceeding the value x = 30. For this reason, the value of x=30 may be identified as most optimal in terms of comparing the optimal length of the road and the time of its acquisition. Value x=100 gave the shortest calculation time, but the path was relatively long. The situation is quite the opposite in the case of x=10 when the route was short but the time of calculation was disproportionately elongated.

Table 7
Results obtained for visibility graph for Series 2

2	n/a	00:01,3	1756

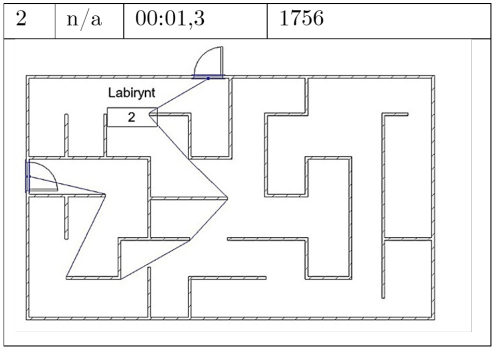


Figure 6
Visual comparison of results for Series4

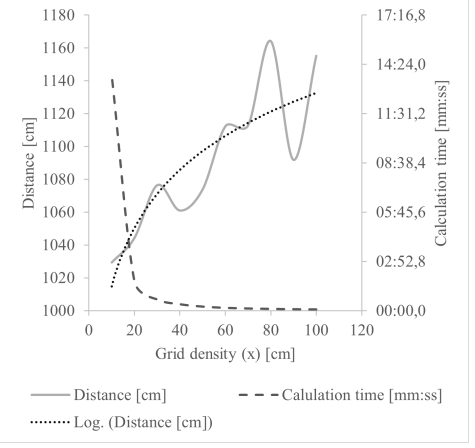
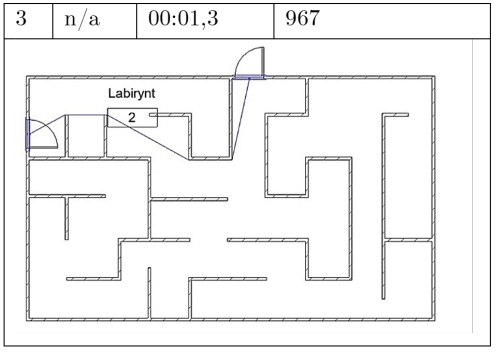


Table 8
Results obtained for visibility graph for Series 3

3	n/a	00:01,3	967



With the visibility method, the route is known to be the shortest because it connects in a straight line key decision points where the change of direction takes place (Tab. 6-9).

4	n/a	00:01,4	2279

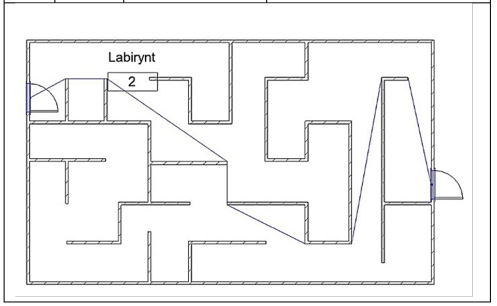


Table 9
Results obtained for visibility graph for Series 4

		Series 1		Series 2	
Method	Grid –x [cm]	Calc. time [mm:ss]	Dist. [cm]	Calc. time [mm:ss]	Dist. [cm]
Matrix method	100	00:05,0	1975	00:05,2	2153
	30	00:45.6	1943	00:46.3	2022
	10	15:39,1	1856	16:01,2	1872
Vis. graph	n/a	00:01,1	1778	00:01,3	1756

Table 10
Comparison of results obtained for matrix method (x=100, x=30, x=10) and visibility graph for Series 1-2

		Series 3		Series 4	
Method	Grid –x [cm]	Calc. time [mm:ss]	Dist. [cm]	Calc. time [mm:ss]	Dist. [cm]
Matrix method	100	00:04,4	1155	00:04,5	2719
	30	00:39.4	1076	00:42.1	2512
	10	13:43,8	1029	13:54,5	2421
Vis. graph	n/a	00:01,3	967	00:01,4	2279

Table 11
Comparison of results obtained for matrix method (x=100, x=30, x=10) and visibility graph for Series 3-4

In the comparison of the calculation times, it can be seen that they are even shorter compared to the value of x = 100 for the matrix grid (Tab. 10-11).

In order to confirm the findings, a series of further tests were conducted. For the comparison, x = 30 was chosen as the optimal choice for the matrix version and x = 100 as the variant of the shortest computation time.

This supports the hypothesis that in the matrix method the measurement time is proportional to the scale of the space under consideration. The number of points that must be computed is directly proportional to the size of the space. This implies that discovering all possible variations is the most computationally challenging activity of the script deciding the shortest path. The changing surface does not impact the performance of the visibility method as it is depending on the number of vertices and not the distance between them.

To validate the above-mentioned hypothesis, additional maze-complicating walls were applied to the Series 5 floorplan in Series 6. (Tab. 14-15).

The length of the road rose more than twice, resulting in a substantial improvement in the route's width and measurement time. The increase in complexity of the room outline has a direct impact on a number of connections between edges in the visibility graph, but the objective time extension is insignificant compared to the matrix method.

Table 12
Results obtained for matrix method (x=30) for Series 5

5	30	01:29,0	2510

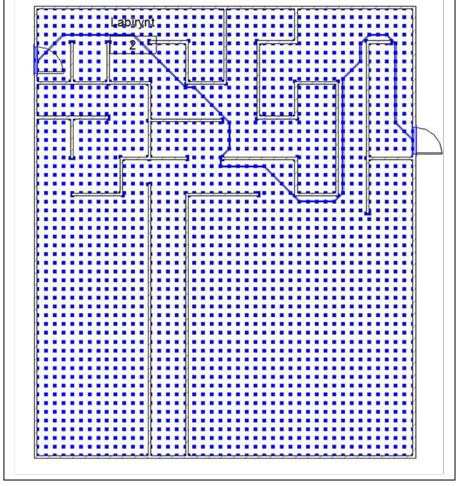


Table 13
Results obtained for visibility graph for Series 5

5	n/a	00:01,3	2279

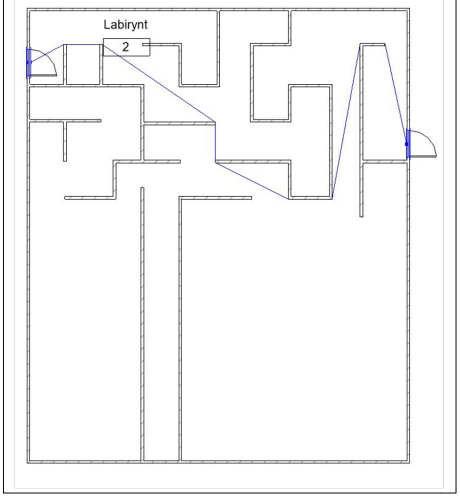
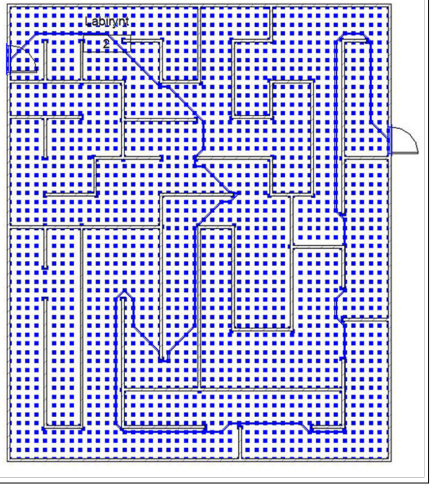


Table 14
Results obtained for matrix method (x=30) for Series 6

6	30	02:13,7	5338



CONCLUSIONS

As mentioned in the Introduction chapter, the study aims to broaden the knowledge of the possible application of the BIM model throughout the building life cycle. By examining further potential applications of this model, the chances of further use of BIM documentation prepared for construction are increased and the attractiveness of using BIM in existing buildings increases.

The findings of research into identifying the shortest path between points A and B using the point matrix in the BIM model are presented in this article. The predicted increase in computing time with increasing mesh density, as stated in the literature, was positively verified (Chen et al., 2018; Hahmann et al., 2018). The relationship between the mesh density and the efficiency of the matrix method is not linear and attempts can be made to indicate the optimal relationship between the accuracy of the route and the calculation time. The second of the tested methods, the visibility method, turned out to be more efficient both in terms of route length and calculation time.

6	n/a	00:02,2	4888

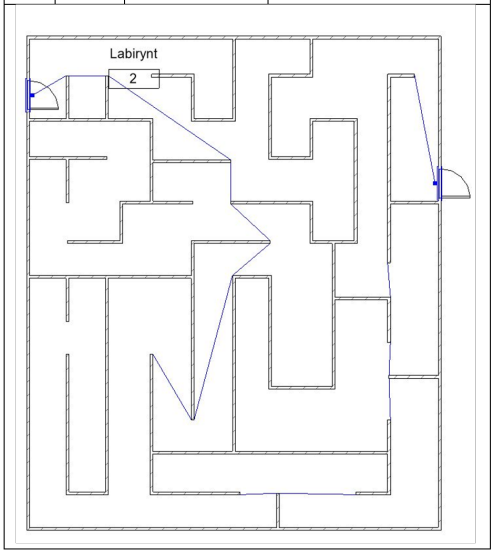


Table 15
Results obtained for visibility graph for Series 6

For this reason, it is a good method to implement in a tool that allows users to search for the shortest route between rooms in the entire building. The already mentioned adaptation of the route to be traveled by a man should be implemented, by assuming an appropriate distance from the wall in decision points.

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